

Transcriptomic and Epigenetic Responses to Flame Retardant Exposure in an Embryonic Cell Model

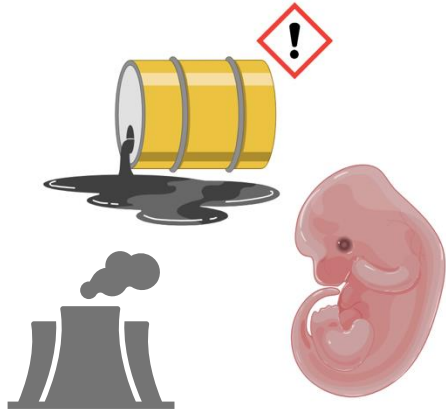
Logan Germain, Lihua Xue & Louise M. Winn

*Department of Biomedical and Molecular Sciences,
Queen's University*

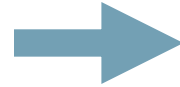
**DENVER
COLORADO**

Abstract Title: Investigating differentially methylated regions of metabolic genes in aquatic embryonic cells following exposure to the flame retardant triphenyl phosphate using whole genome bisulfite sequencing

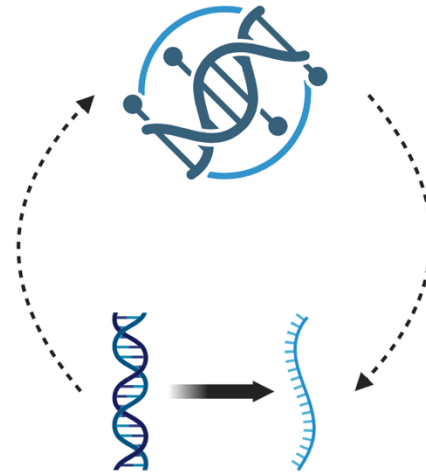
- Pharmaceuticals
- Maternal and fetal conditions
- Environmental pollutants



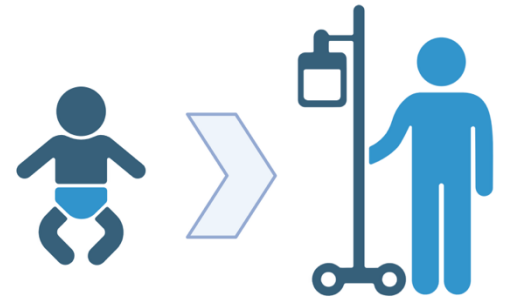
Exposure During
Development
(**window of
susceptibility**)



Epigenome Changes



Transcriptome Changes



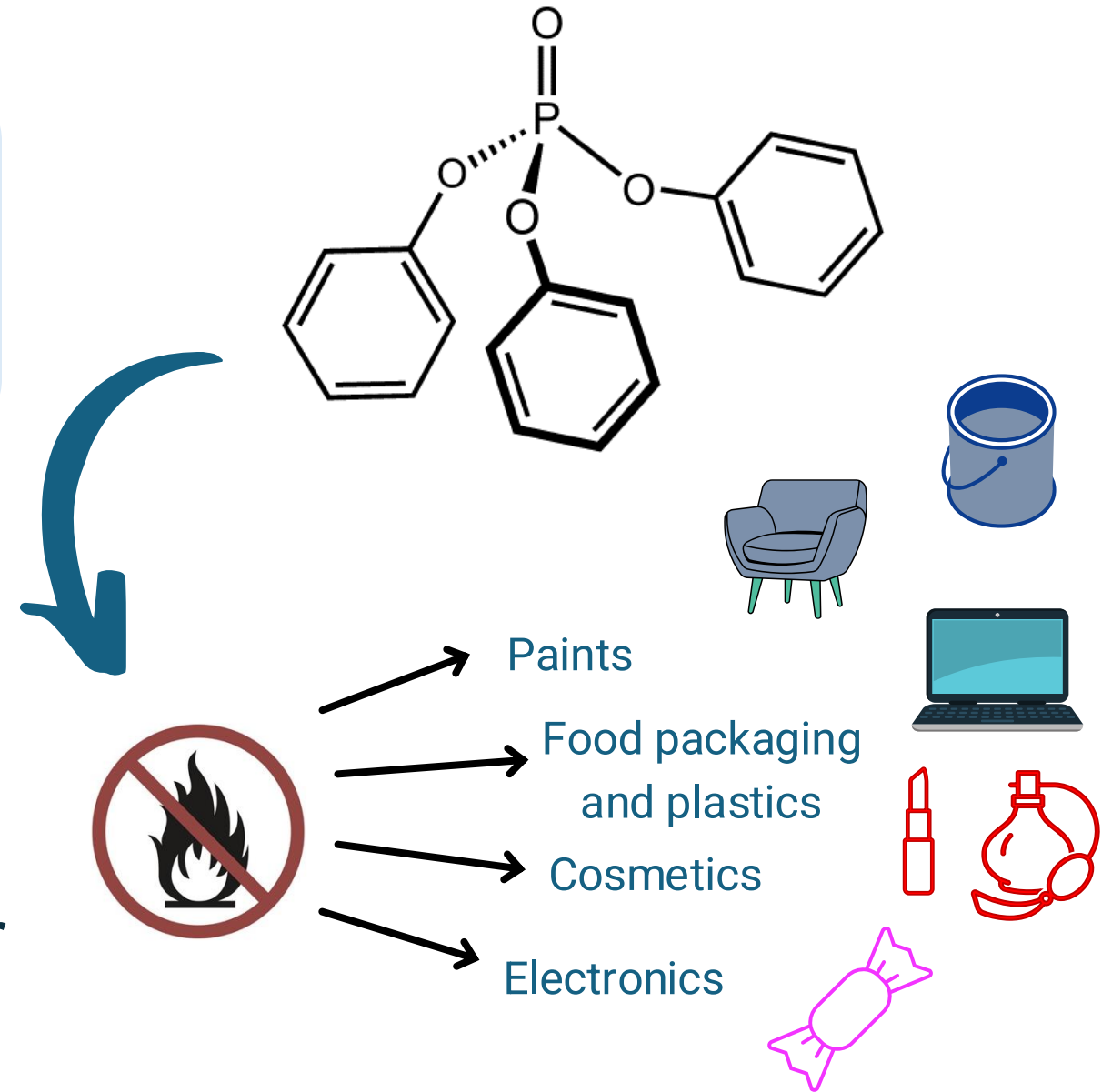
Disease or Dysfunction

(Chen et al., (2023), Cao et al., (2025), Zhou et al., (2019), Bui et al., (2024), Xu et al., (2024))



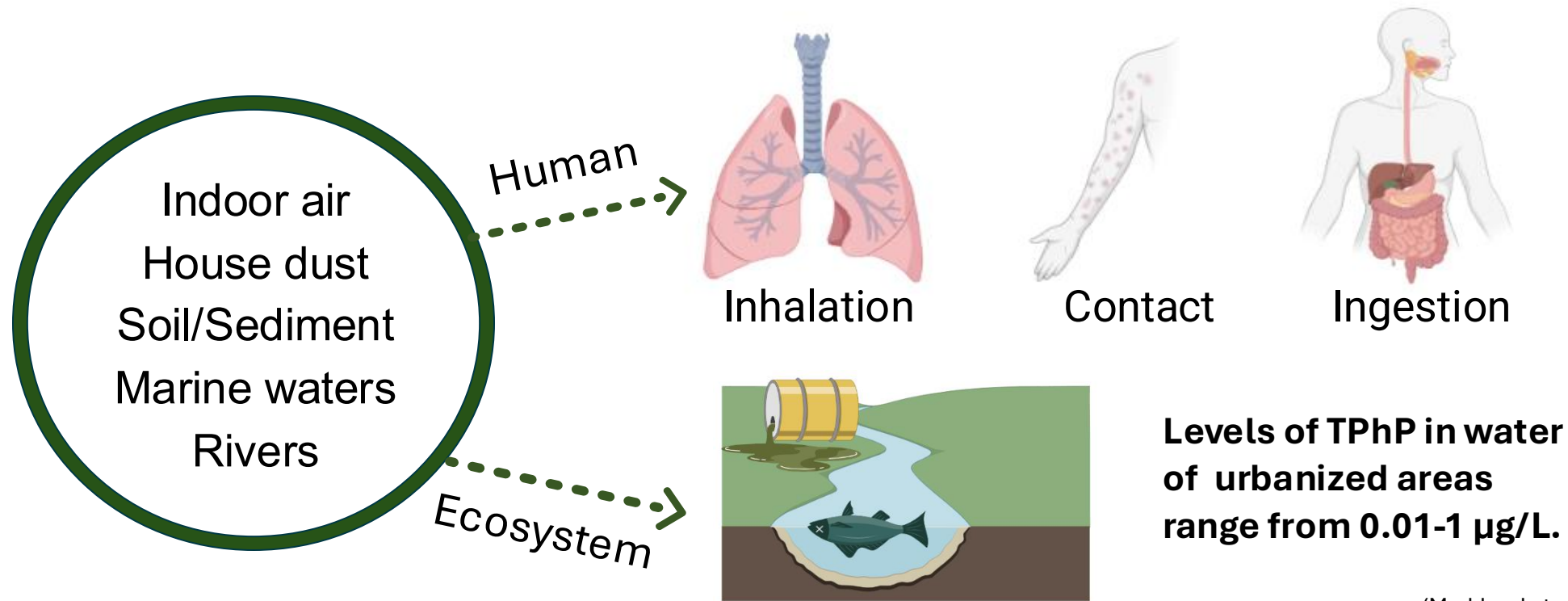
triphenyl phosphate (TPhP)

- Organophosphate flame retardant and plasticizer
- Used in a variety of consumer and industrial products
- **Not currently regulated in Canada or US, with restrictions in the EU**



EXPOSURE

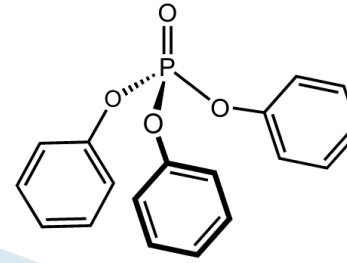
- TPhP is not chemically adhered to materials – often added as a powder.
- Released into surrounding environments via volatilization.
- **Found present in aquatic, terrestrial and atmospheric environments.**



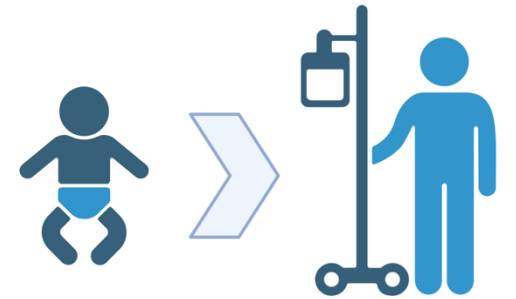
(Marklund et al., 2005; Sundkvist et al., 2010)



Triphenyl phosphate (TPhP):
Organophosphate flame
retardant and plasticizer



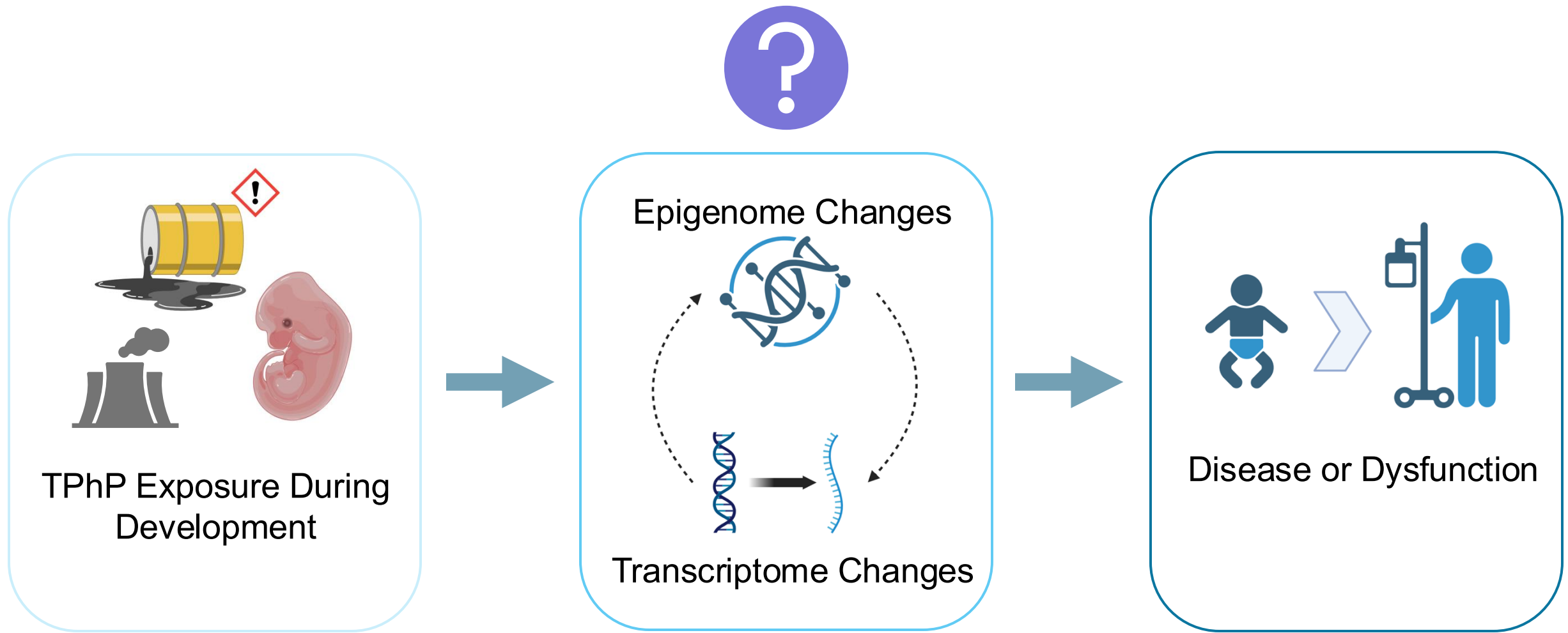
- Metabolic Syndromes
- Endocrine Disruption
- Neurotoxicity
- Oxidative Stress
- Cancer Risk



Disease or Dysfunction

(Chen et al., (2023), Cao et al., (2025), Zhou et al., (2019), Bui et al., (2024), Xu et al., (2024))





(Chen et al., (2023), Cao et al., (2025), Zhou et al., (2019), Bui et al., (2024), Xu et al., (2024))



Making a Case for the Fish Model



Fish are powerful alternative models in vertebrate developmental biology

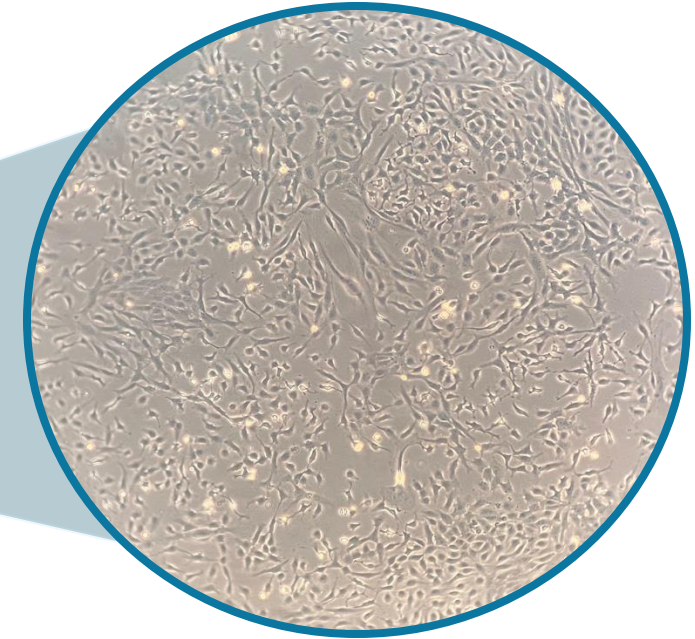
- Extremely sensitive to their environments – “Canaries in the coal mine”
- Many developmentally-important genes are highly conserved between fish, humans and other vertebrates.
- **Embryos are cost-effective, ethically accessible, and easier to work with at scale**



Rainbow Trout



STE-137



STE-137: An Immortalized Embryonic Cell Line

- Derived from pooled rainbow trout embryos; first established in the 1970s
- Our lab is the first to apply STE-137 in toxicological screening

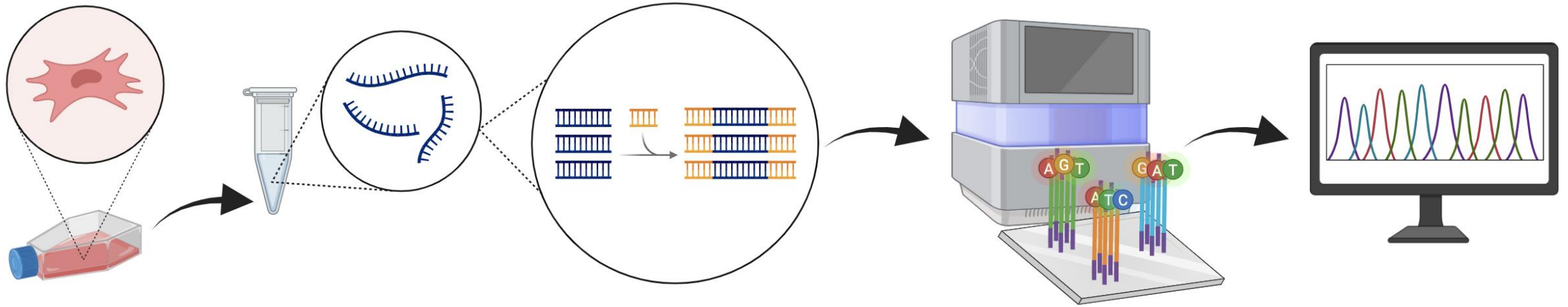


Transcriptomic Impacts of TPhP

Goal: Identify transcriptional changes in STE-137 cells following TPhP exposure to uncover disrupted biological pathways.



RNA-seq METHODS



- 24h Exposure to 80 μ M TPhP or Control (0.08% DMSO)
- n = 9-10 per group

In Collaboration with the Queen's University Cardiopulmonary Unit

1) Exposure & RNA Extraction

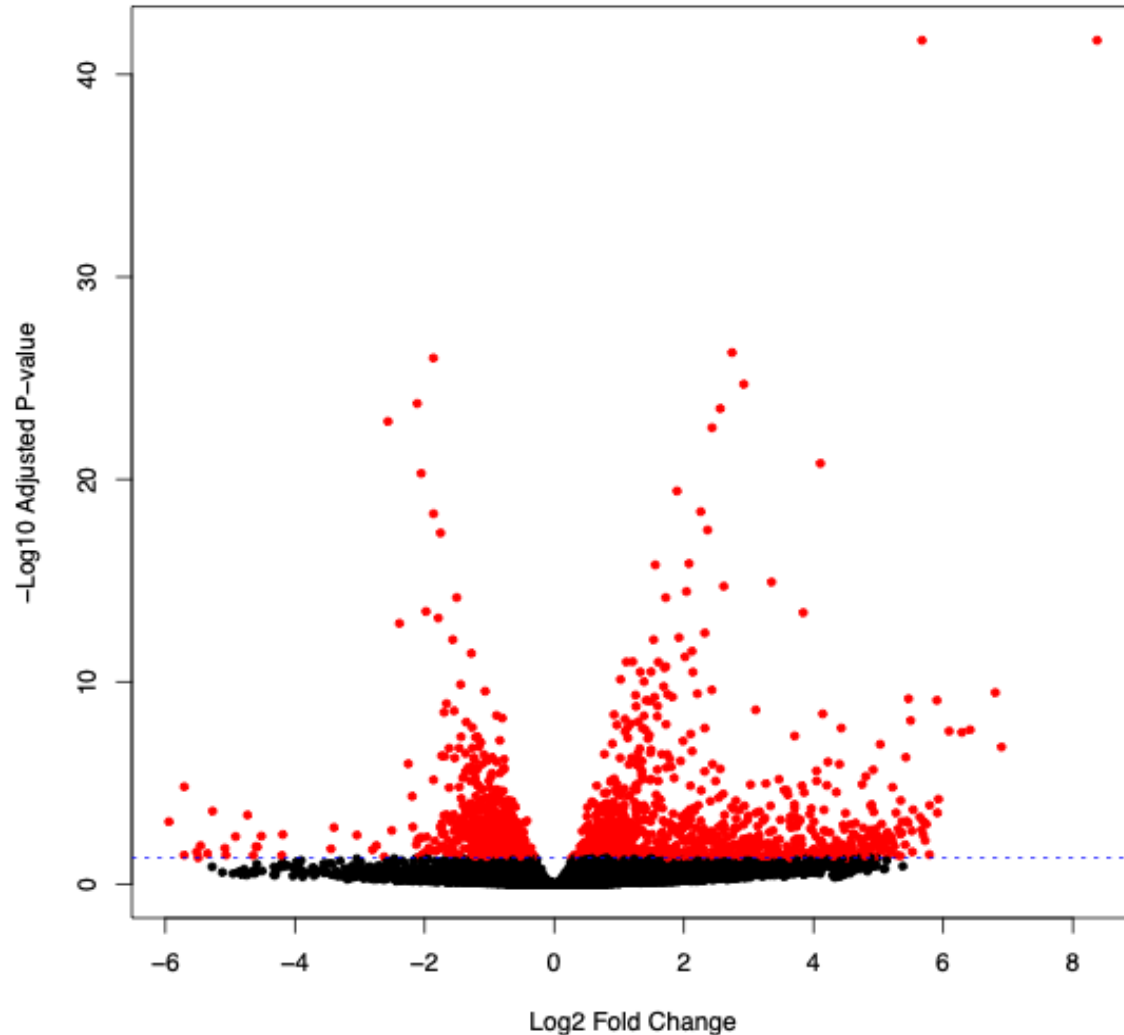
2) Library Preparation

3) Illumina Sequencing

4) Data Processing & Analysis



Volcano Plot of Differentially Expressed Genes



The transcriptome is significantly altered by TPhP exposure

1622 Differentially Expressed Genes (DEGs) control vs. exposed group.

663 DEGs are downregulated.

959 DEGs are upregulated.



Most Significant DEGs (Top 10)

Gene Name	Gene Description	Log2 Fold Change	Adjusted p-value
soat2	sterol O-acyltransferase	8.38511081	4.96E-42
cidec	cell death inducing DFFA like effector c	5.79663856	6.75E-38
mcm2	minichromosome maintenance complex component 2	-1.912322	3.85E-32
gtpbp2	GTP-binding protein 2	3.01979136	7.47E-29
ddit3	DNA-damage-inducible transcript 3	2.82178335	8.59E-29
ywhag1	3-monooxygenase/tryptophan 5-monooxygenase activation protein, gamma polypeptide 1	2.67009259	2.16E-28
LOC110501449	cell division cycle associated 7	-2.625401	2.56E-28
mcm3	minichromosome maintenance complex component 3	-2.1217417	1.44E-24
map1ab	microtubule-associated protein 1Ab	1.94454894	2.99E-23
mcm5	minichromosome maintenance complex component 5	-2.091256	1.43E-22



Most Significant DEGs (Top 10)

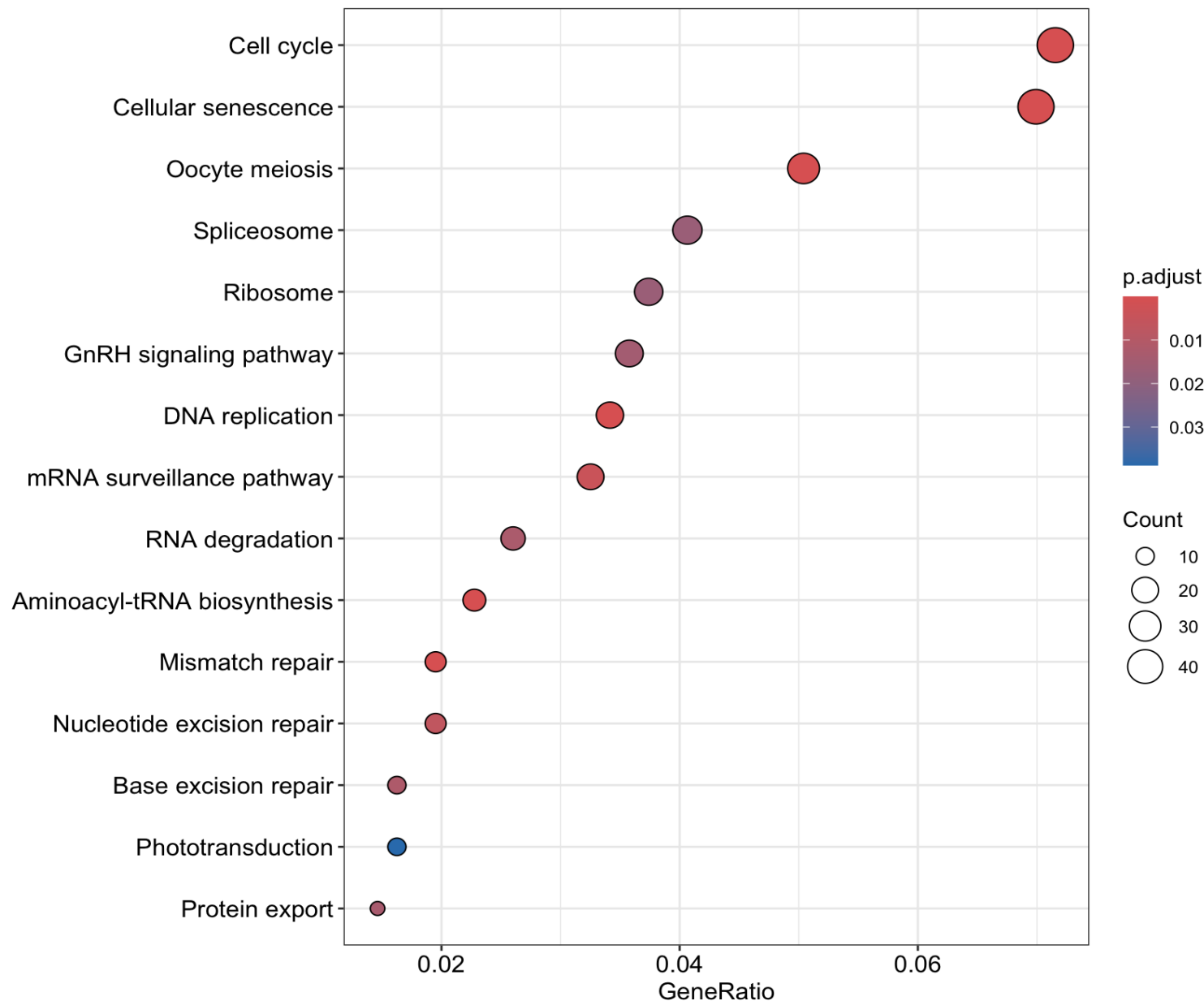
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Most Significant DEGs (Top 10)

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soat2	sterol O-acyltransferase	8.38511081	4.96E-42
cidec	Sterol O-acyltransferase: - Catalyzes the formation of cholesterol esters from cholesterol and long-chain fatty acyl-CoA - Maintains cholesterol homeostasis mainly in endoplasmic reticulum of intestinal and liver tissues 300% upregulation reinforces TPhP as a metabolic and endocrine disrupting chemical		6.75E-38
mcm2			3.85E-32
gtpbp2			7.47E-29
ddit3			8.59E-29
ywhag1			2.16E-28
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mcm5	minichromosome maintenance complex component 5	-2.091256	1.43E-22





PATHWAY ANALYSIS

- TPhP exposure disrupts key cellular processes: **DNA replication and repair, mRNA processing, cell cycle progression, and endocrine signaling.**
- **Key Take-Away:** Broad impact on developmentally-important genes, which indications for DNA damage and endocrine disruption.



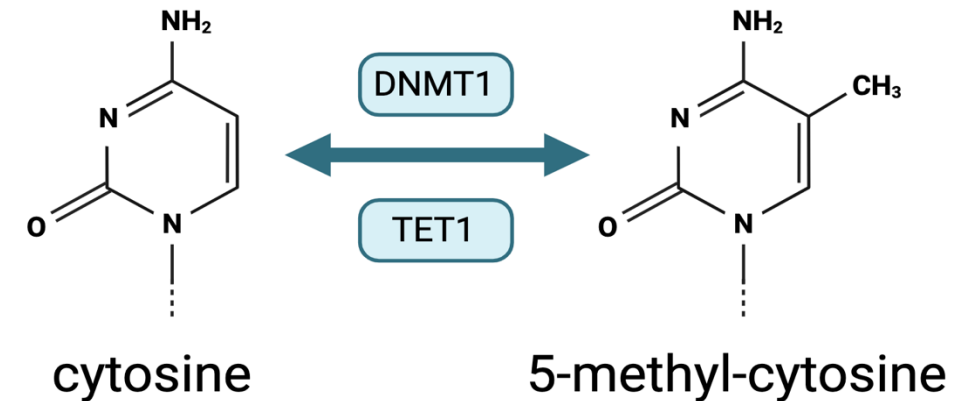
Epigenetic Impacts of TPhP

Goal: Identify **DNA methylation changes across the entire genome in STE-137 cells following TPhP exposure** to identify gene-specific epigenetic alterations.

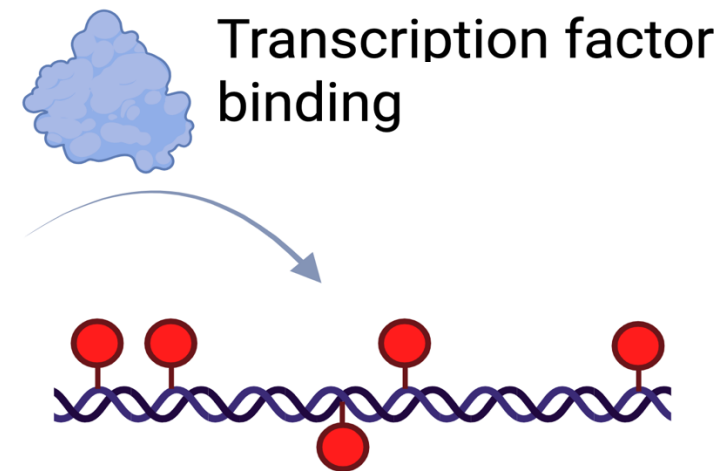


DNA Methylation

An **epigenetic modification** where a methyl group is added to the 5-carbon of cytosine – usually at a **CpG site**.

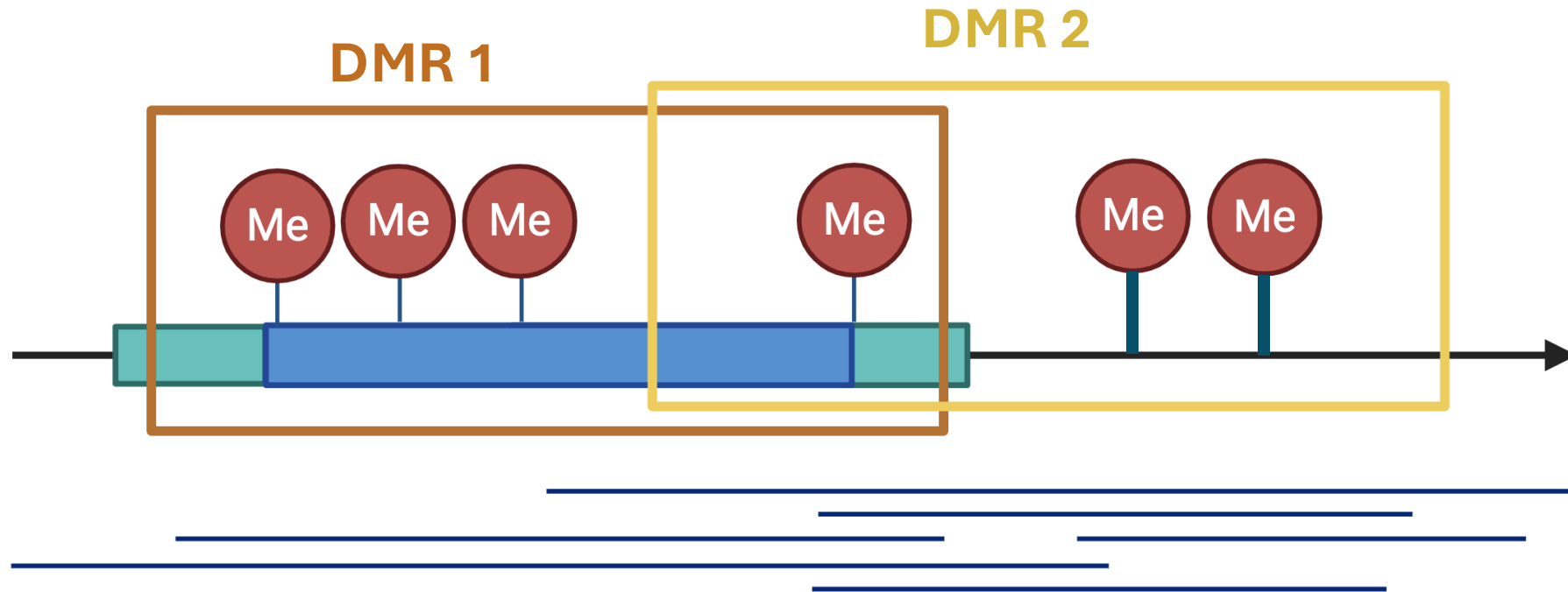


Influences gene expression – can have POSITIVE and NEGATIVE regulation on expression.

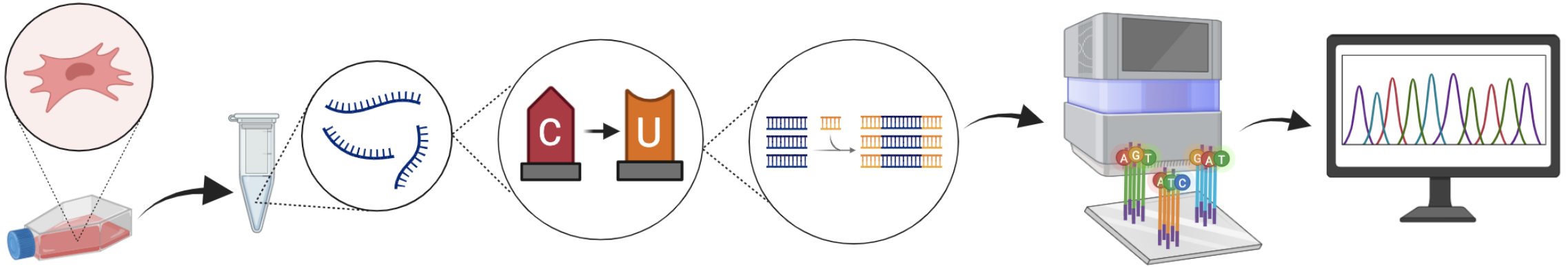


Identifying Localized Methylation Changes

Differentially Methylated Region (DMR): Any **500bp window** across the entire genome with a different **DNA methylation percentage** between control and exposed groups.



Whole Genome Bisulfite Sequencing



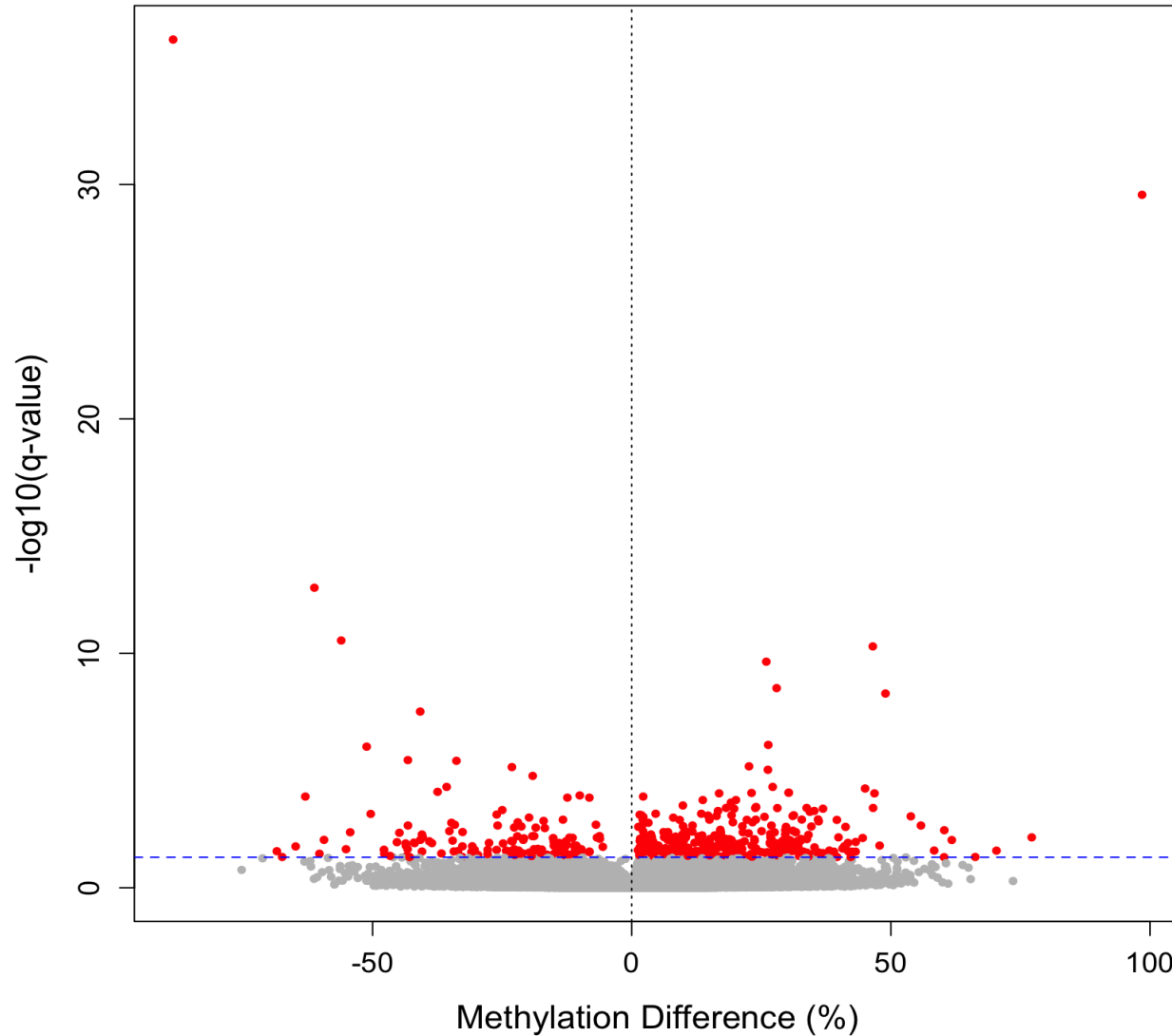
- 24h Exposure to 80 μ M TPhP or Control (0.08% DMSO)
- n = 9-10 per group

In Collaboration with the McGill Genome Centre & Canadian Centre for Computational Genomics

- 1) Exposure & DNA Extraction
- 2) Bisulfite Conversion & Library Prep at McGill Genome Centre
- 3) Illumina Sequencing at McGill Genome Centre
- 4) Data Processing & Analysis



Volcano Plot of DMRs



The epigenome is hypermethylated by TPhP exposure.

This may have profound impacts on gene regulation.

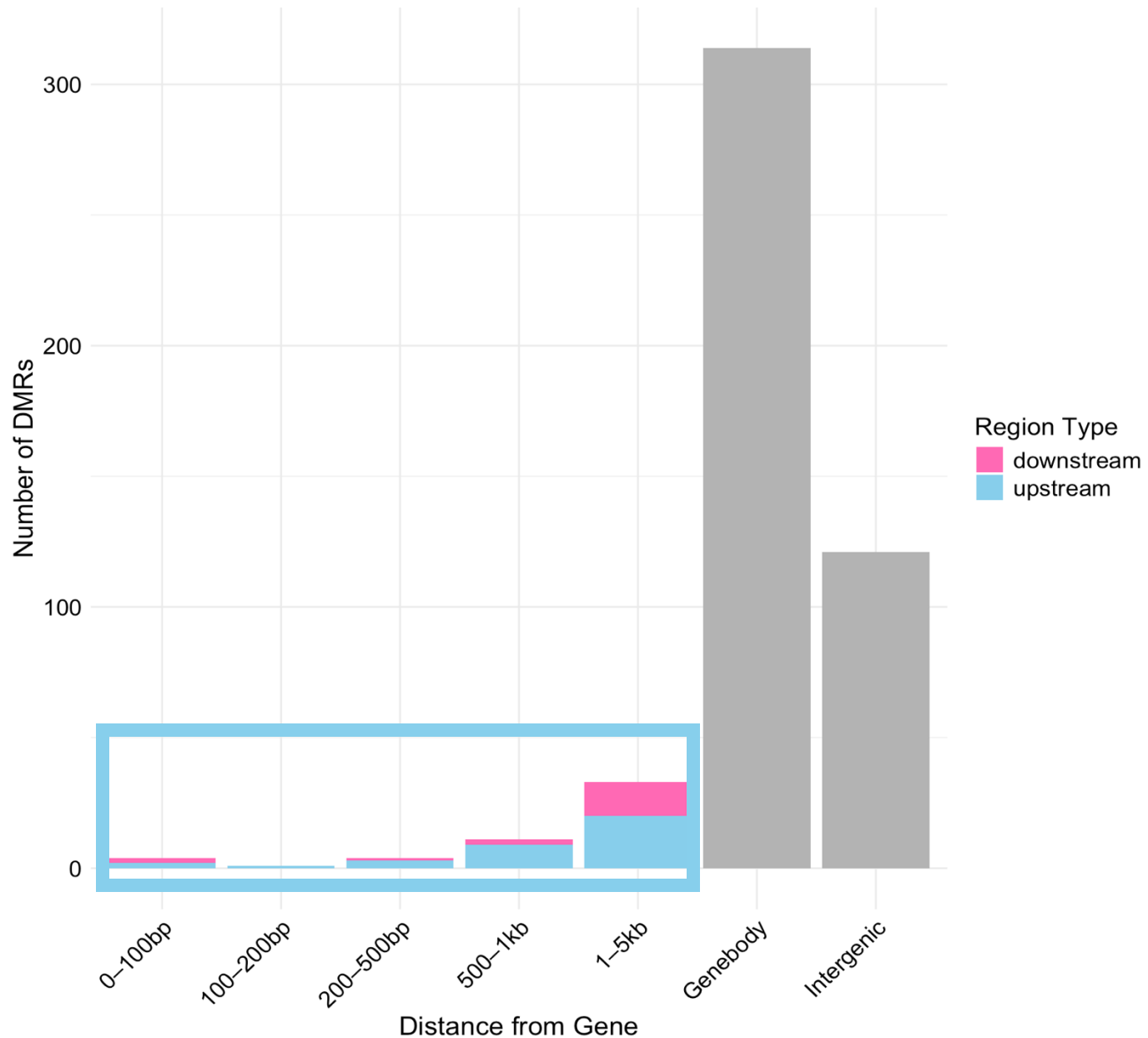
488 significant DMRs

345 hypermethylated (70.7%)

143 hypomethylated (29.3%)

$-\log_{10}1.3 = \text{q-value} < 0.05$

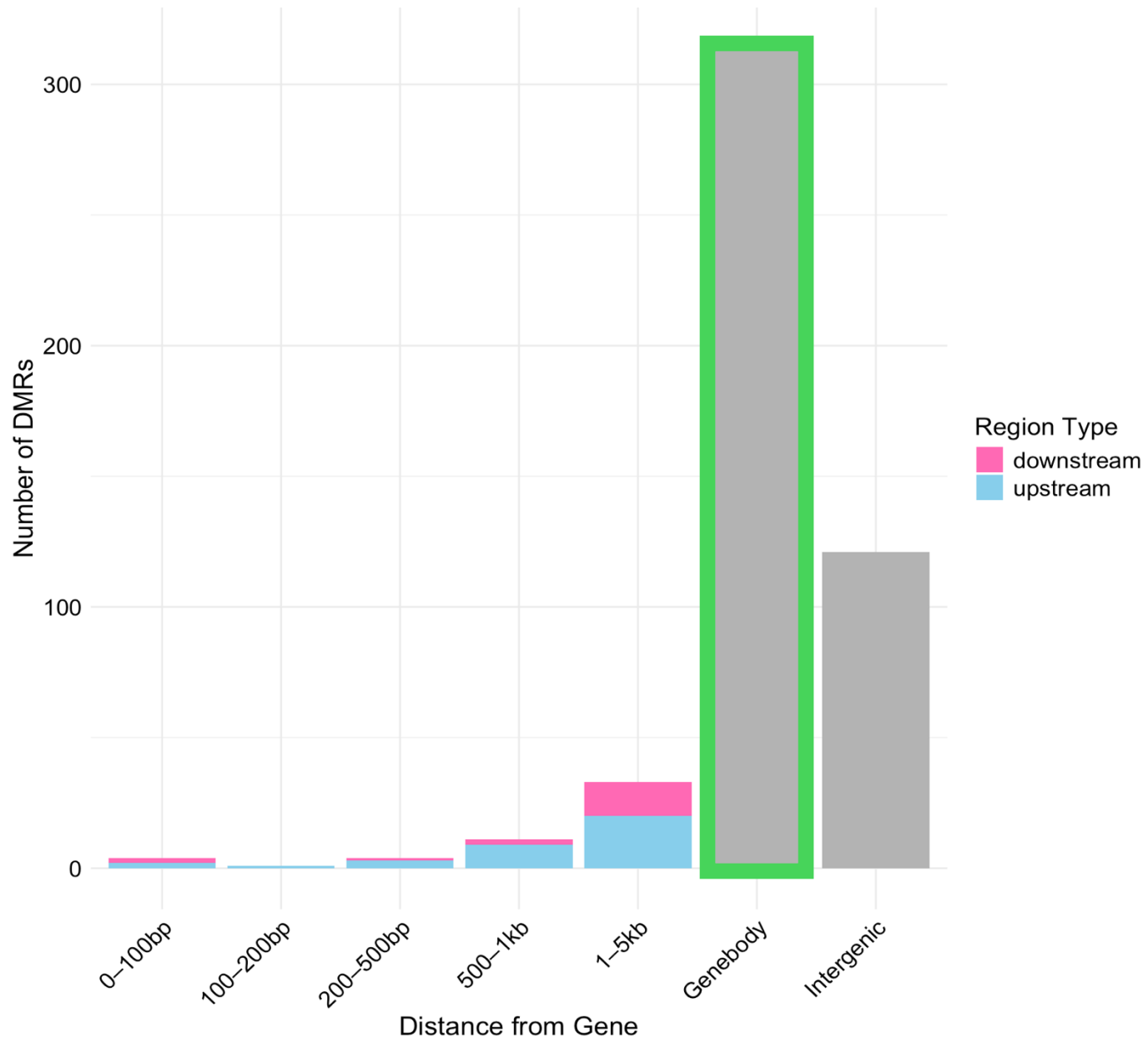




DMR Locations

- **7% are within a suspected promoter region (<5000bp from TSS).**



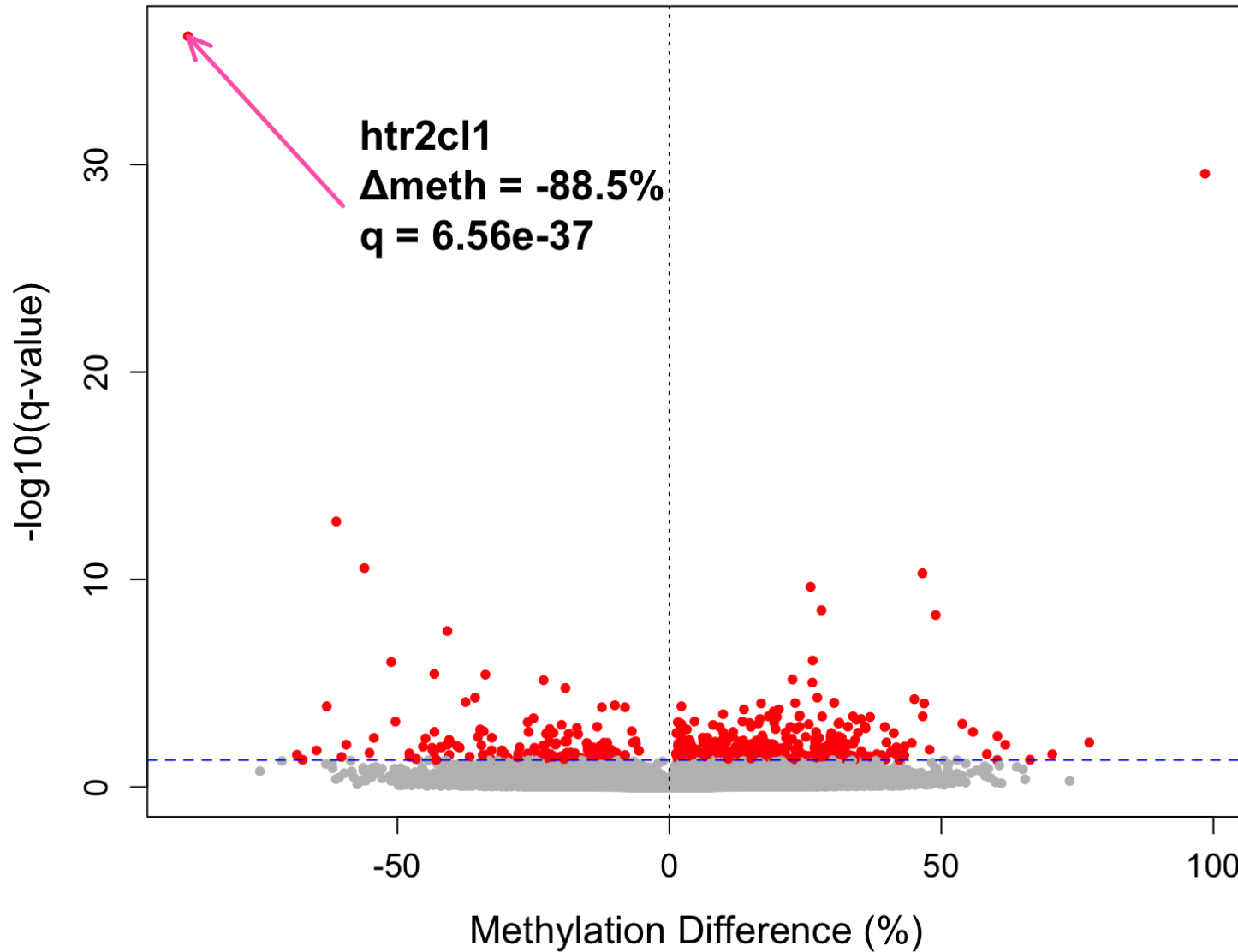


DMR Locations

- 7% are within a suspected promoter region (<5000bp from TSS).
- **64% overlap with a known gene body.**



Volcano Plot of DMRs



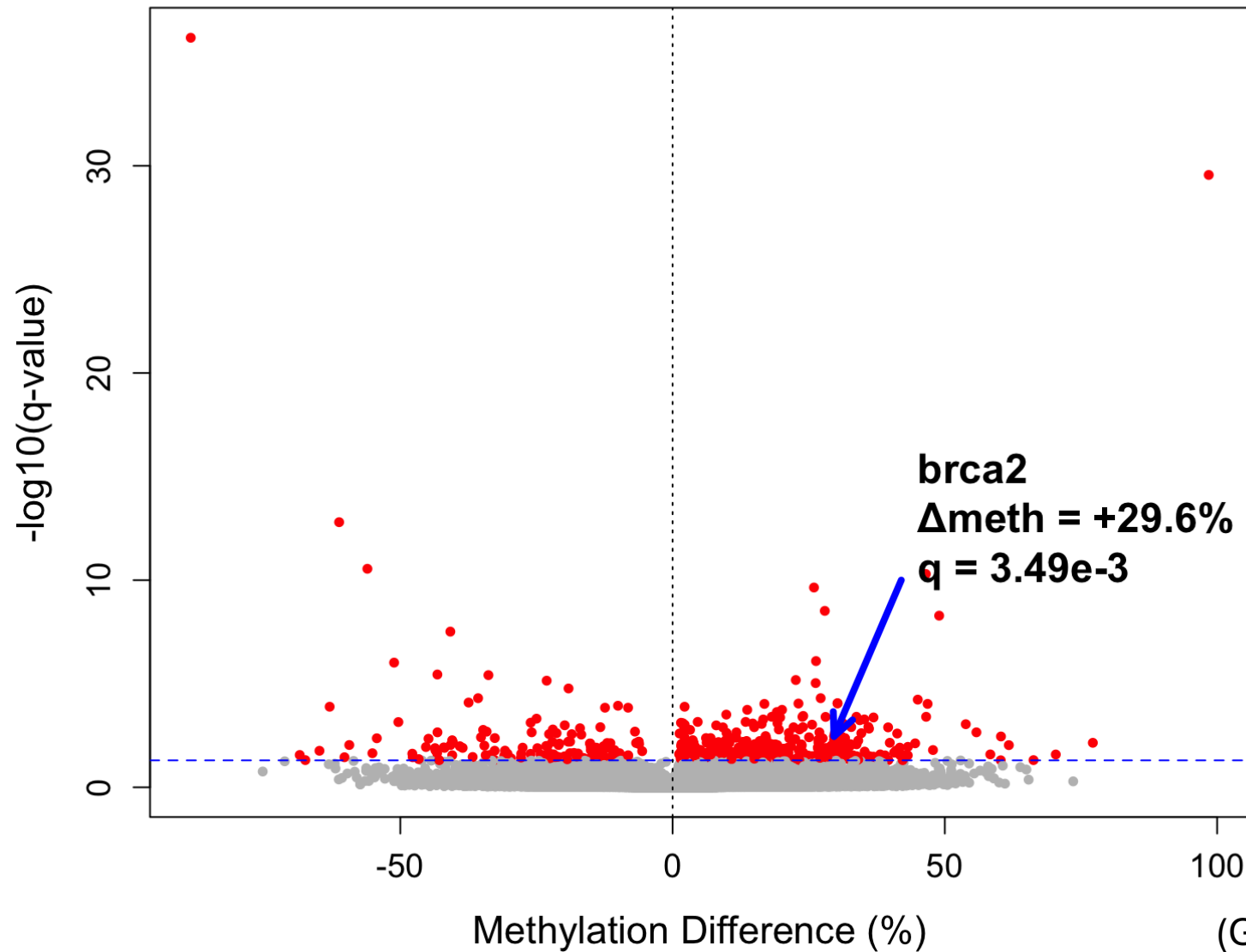
5-HT_{2C} (serotonin) Receptor:

- Encodes a G-protein coupled serotonin receptor
- Involved with downstream **serotonin neurotransmitter signaling pathway**:
 - Neurological development, anxiety, behaviour and obesity

(Reynolds et al., 2005; Pannanen et al., 2013; ZFIN)



Volcano Plot of DMRs



Breast Cancer Gene 2:

- Involved with DNA break repair
- Main task as a **tumour suppressor gene**
- Hypermethylation shown to be predictive of poor breast cancer prognosis in humans

(Gorodetska et al., 2019; Panagopoulou et al., 2024)

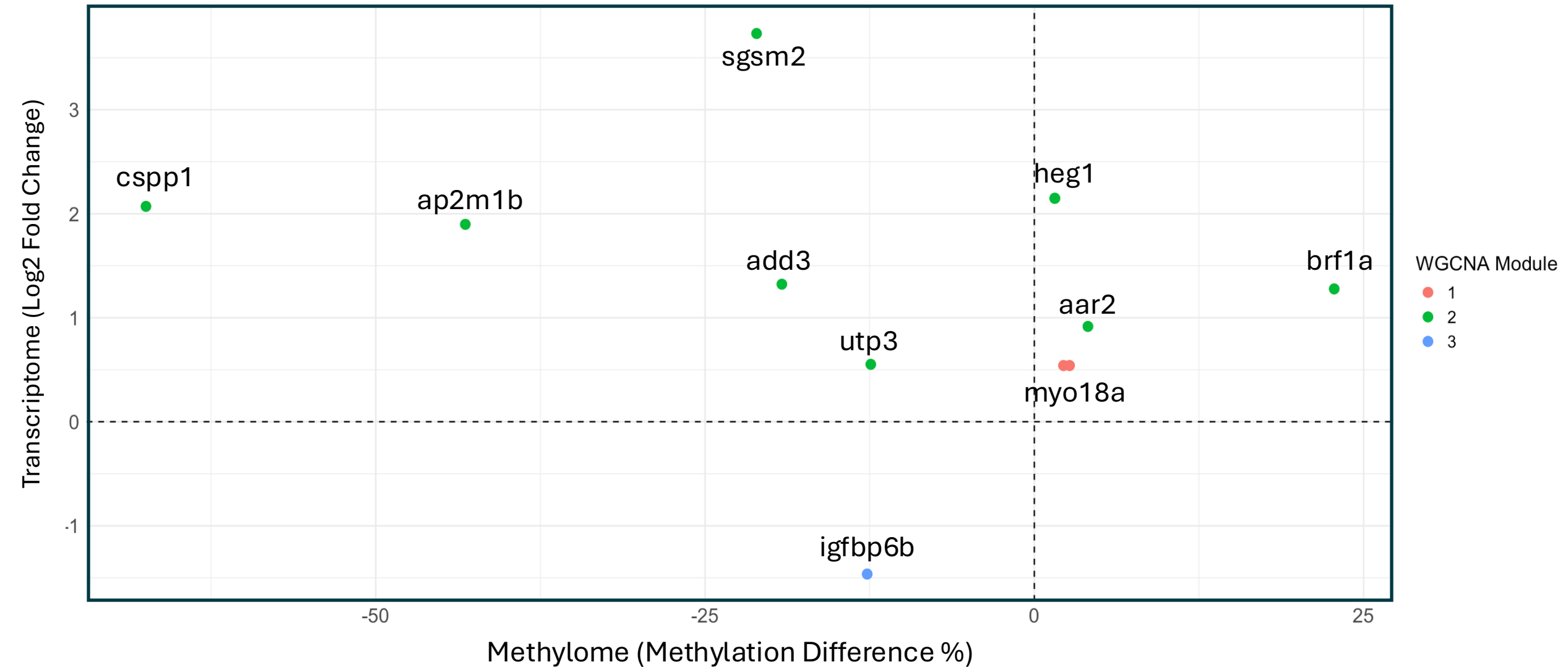


Multi-Omic Integration

Goal: Identify genes that have both transcriptional and epigenetic disruptions to identify coordinated alterations by TPhP.



Multi-Omic Integration of Transcriptome and Methylome Changes



Multi-Species Comparison

Goal: Utilize publicly available datasets to determine if STE-137 identified the same orthologous pathways disturbed in other species/models to increase confidence in biological relevance.



► *Toxicol Sci.* 2023 Nov 6;197(2):170–185. doi: [10.1093/toxsci/kfad114](https://doi.org/10.1093/toxsci/kfad114) 

Toxicological mechanisms and potencies of organophosphate esters in KGN human ovarian granulosa cells as revealed by high-throughput transcriptomics

[Xiaotong Wang](#)¹, [Andrea Rowan-Carroll](#)², [Matthew J Meier](#)³, [Andrew Williams](#)⁴, [Carole L Yauk](#)⁵, [Barbara F Hales](#)⁶, [Bernard Robaire](#)^{7,8,✉}

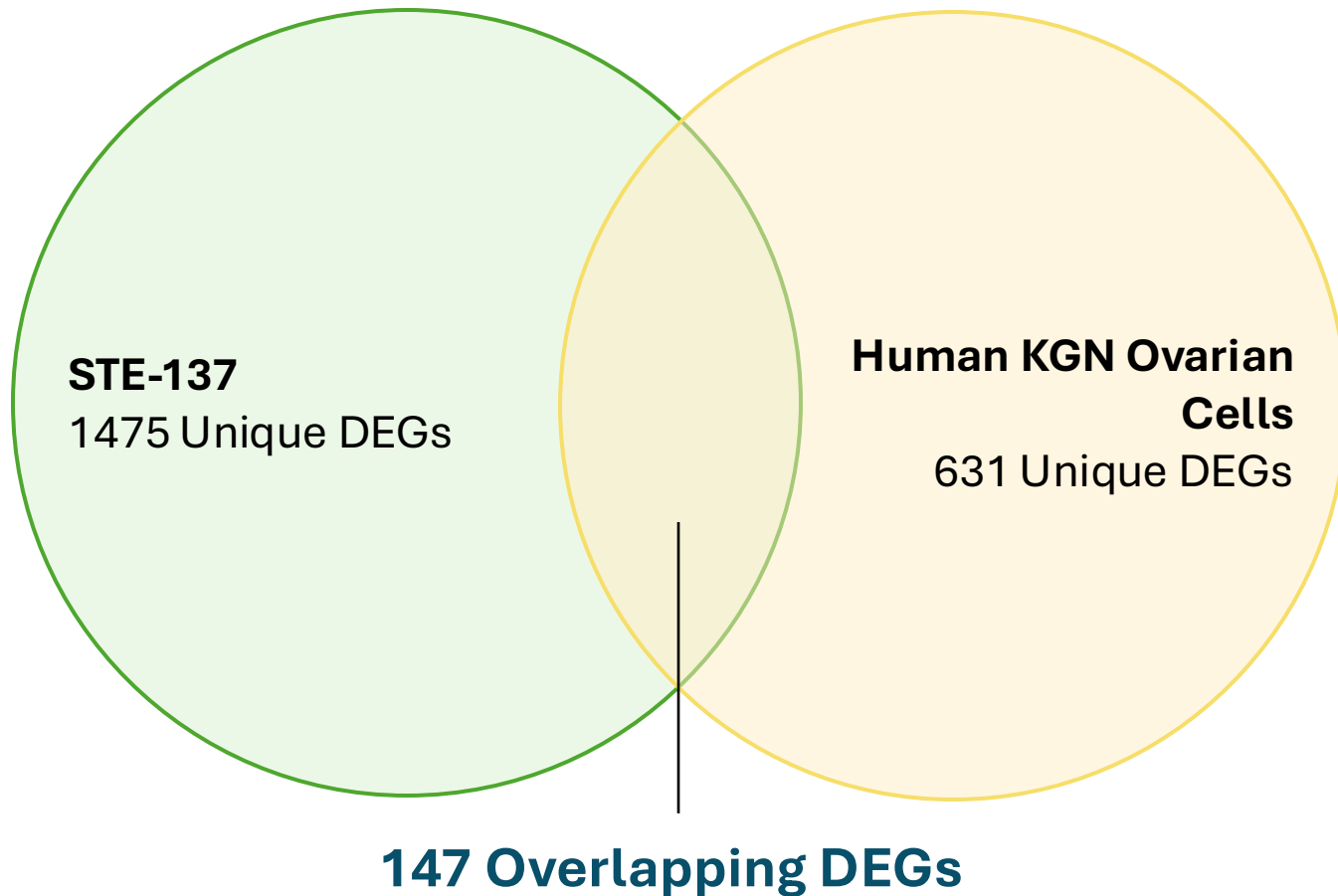
► [Author information](#) ► [Article notes](#) ► [Copyright and License information](#)

PMCID: PMC10823774 PMID: [37941476](#)

- Utilizing publicly available RNA-seq data for **KGN human ovarian granulosa cells** exposed to TPhP, I **re-analyzed their data and identified overlapping DEGs between their study and my study.**
- **Experimental Design:** 50 μ M TPhP or 0.5% DMSO (Control) for 48h.



Overlapping Orthologous Transcriptome Impact



Key Take-Away:

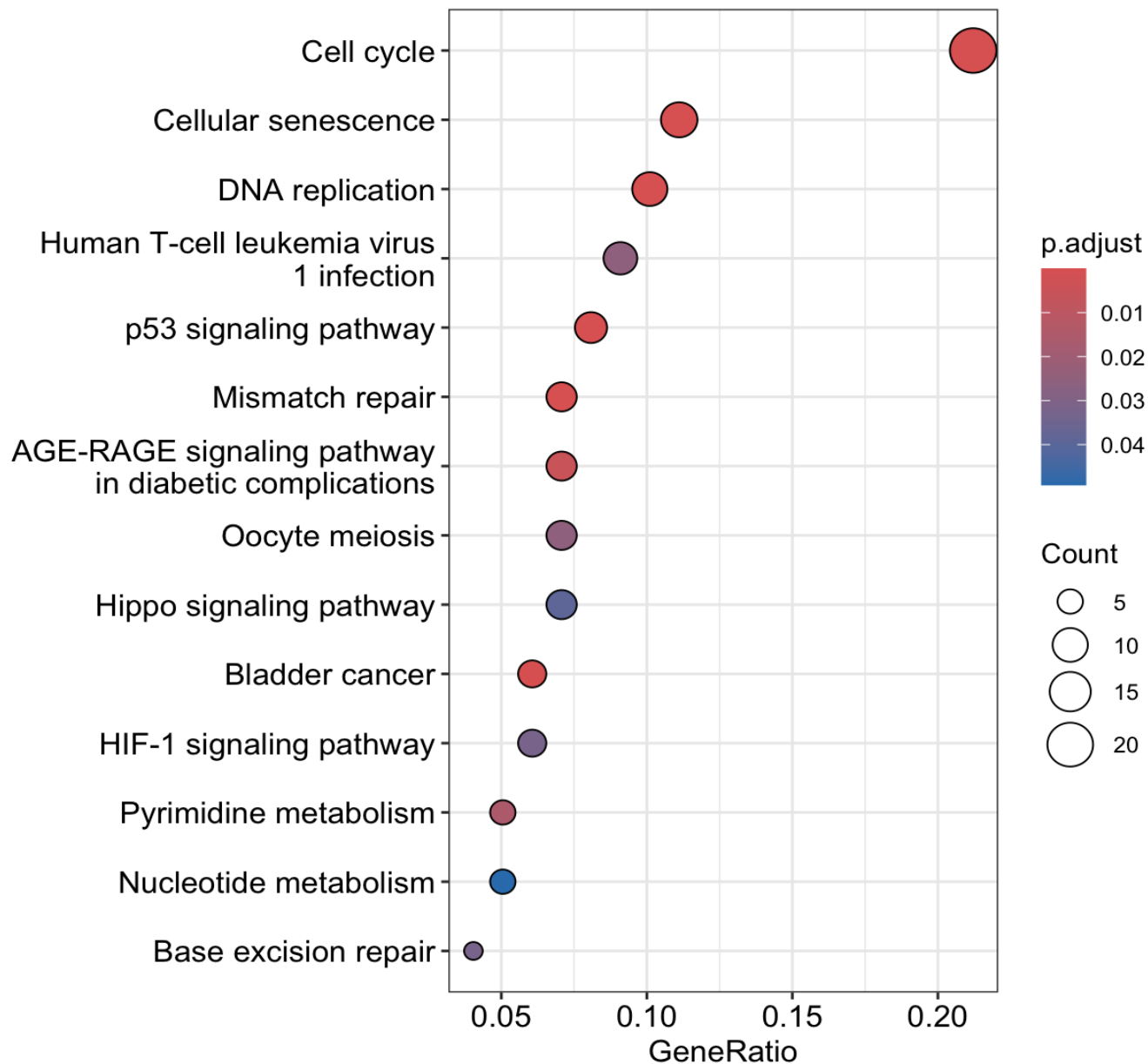
My study identified 1622 DEGs.

Wang et al., (2023) identified 778 DEGs.

147 orthologous DEGs overlap between both studies.

(Wang et al., 2023)



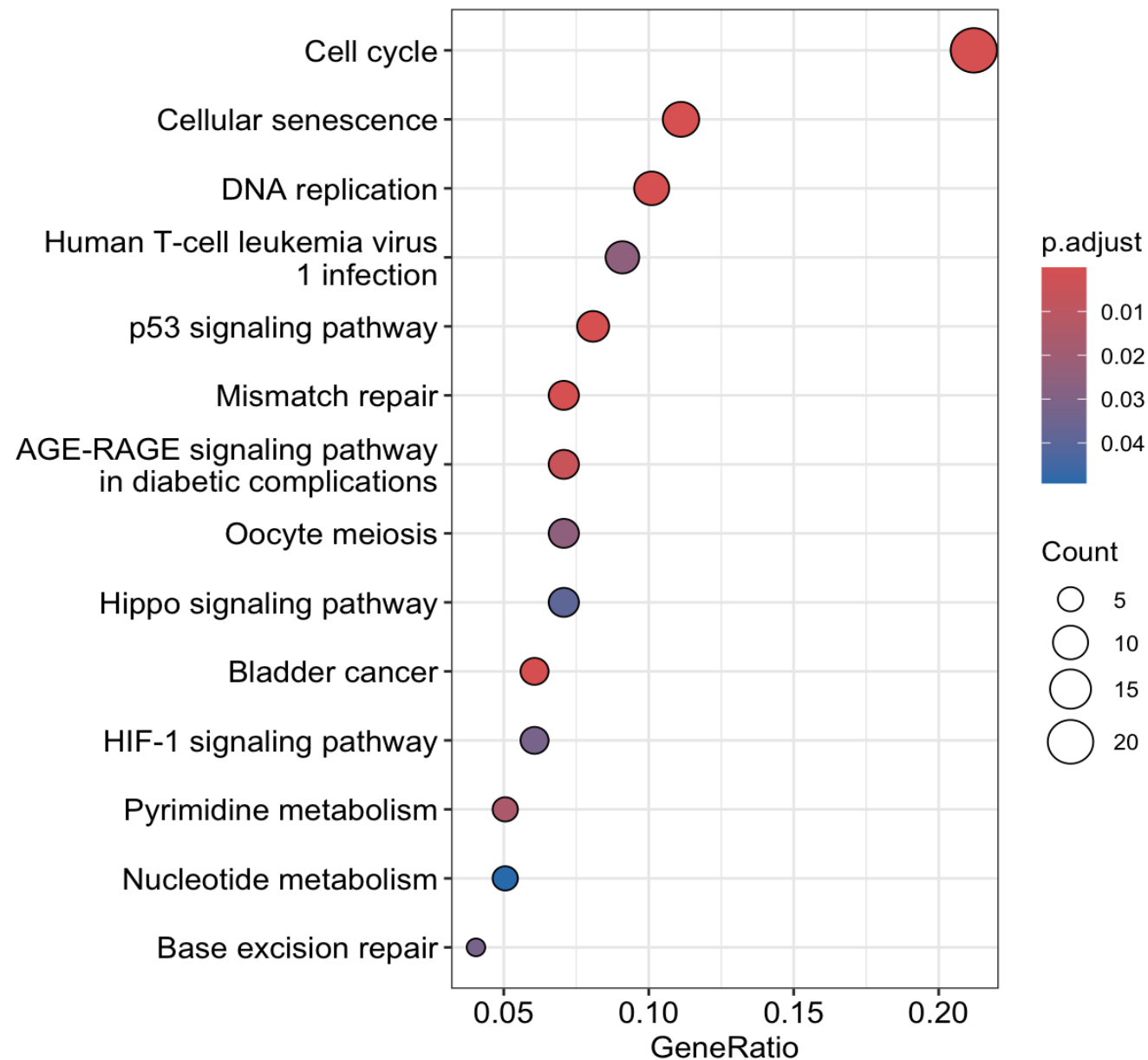


Pathway Analysis

Overlapping DEGs reveal disruptions to:

- **Cell cycle**
- **Cellular senescence**
- **DNA fidelity and repair**
- **Metabolism and endocrine-related pathways**





My model (STE-137) was able to identify these same pathways that were also disturbed in humans.

Increases confidence in its ability to act as a toxicological screening tool!



Key Conclusions

- TPhP exposure significantly **alters the transcriptome of embryonic cells.**
 - DNA repair
 - Cell cycle control
 - Metabolism
 - Endocrine signaling

Key Conclusions

- TPhP exposure significantly alters the transcriptome of embryonic cells.
- **TPhP exposure led to epigenetic reprogramming in embryonic cells.**
 - Particularly, hypermethylation within gene bodies and regulatory regions.

Key Conclusions

- TPhP exposure significantly alters the transcriptome of embryonic cells.
- TPhP exposure led to epigenetic reprogramming in embryonic cells.
- Multi-omic integration revealed **10 genes with both differential expression and methylation changes**.
 - Regulators of growth signaling (e.g., *igfbp6b*, *brf1a*)
 - Cytoskeletal structure (*add3*, *cspp1*)
 - RNA processing (*aar2*, *utp3*)
 - Cell-cell adhesion, cardiovascular development (*heg1*)

Key Conclusions

- TPhP exposure significantly alters the transcriptome of embryonic cells.
- TPhP exposure led to epigenetic reprogramming in embryonic cells.
- Multi-omic integration revealed **10 genes with both differential expression and methylation changes**.
- Cross-species comparisons between trout embryonic and human ovarian cells shows **147 genes with conserved transcriptional effects**.
 - STE-137 as a New Approach Methodology (NAM) for toxicological screening?

Key Conclusions

- TPhP exposure significantly alters the transcriptome of embryonic cells.
- TPhP exposure led to epigenetic reprogramming in embryonic cells.
- Multi-omic integration revealed **10 genes with both differential expression and methylation changes**.
- Cross-species comparisons between trout embryonic and human ovarian cells shows **147 genes with conserved transcriptional effects**.
- Early-life exposure to TPhP may lead to disease and dysfunction phenotypes through **epigenetic and transcriptomic mechanisms**.

Thanks for listening! 😊

Questions? Collaborations? Reach out to me!

Logan Germain

PhD Candidate, Queen's University

15lsg@queensu.ca

ACKNOWLEDGEMENTS

Supervisor & Committee Members:

Dr. Louise Winn

Dr. Christopher Nicol

Dr. Maria Aristizabal

Dr. Charles Hindmarch

Winn Lab Members:

Lihua Xue

Lauren Brown

Megan Cull

Perri Grant

Delaine Pereira

Katryna Seabrook

Holly Mackay



The Winn Lab



Queen's
UNIVERSITY



NSERC
CRSNG



QCPU

Queen's CardioPulmonary Unit

McGILL
GENOME
CENTRE



Canadian Centre for
Computational
Genomics

REFERENCES

Link to Google Document with reference citations:

<https://docs.google.com/document/d/1bss8GHZSWYRx0czaJvlznWhMq2zKc9YlixuSoHuBGuc/edit?usp=sharing>